05 / System architecture

12 Spring Boot services, Angular client, 8 owned MySQL schemas and standalone integrations

Config/discovery links apply across participating services; only representative links shown.

Config Server :8888

Central configuration from bundled repository
Authenticated service access

configuration plane

Angular frontend :4200

Passenger pages + admin workspace
Relative /api requests; dev proxy -> :8080

HTTP / JSON / SSE

Eureka Server :8761

Registration + discovery by service identity
Not a booking database

discovery plane

API Gateway :8080

Routing, CORS, request IDs, aggregated Swagger
Downstream services enforce role + ownership

Gateway public routing fanout (not a service-to-service chain)

APPLICATION SERVICES / TRUSTED INTERNAL CALLS

Auth :8081

Accounts, OTP, Google, JWT sessions
Admin invitation and user status
DB: railway_auth_db

User :8082

Profile + passenger Master List
Stable account ownership
DB: railway_user_db

Train :8083

Stations + coordinates; train metadata
Classes, capacity and fare settings
DB: railway_train_db

Schedule :8084

Ordered stops + day offsets
Timetable distance / chronology
DB: railway_schedule_db

Catalogue + timetable calls (Feign)

Reservation :8085

Search + alternatives; fare quotes
Holds, CNF/WL, tickets, cancellation
Support, inbox, reports + outbox
DB: railway_reservation_db

PNR call

PNR :8086

Unique PNR allocation
Internal reservation dependency
DB: railway_pnr_db

Payment :8087

Orders, signature / provider checks
Captures, reconciliation + refunds
DB: railway_payment_db

Notification :8088

Durable event receipt + deduplication
Email / attachments + retry jobs
DB: railway_notification_db

Payment decisions / refund coordination

Additional gateway routes include PNR / notification endpoints; internal authorization still applies

Chatbot :8089 / Spring AI + WebFlux

Gemini + FAQ RAG (SimpleVectorStore); read-only live service tools
InMemoryChatMemoryRepository + bounded semantic cache
SSE output; guardrails + Resilience4j; process-local state

Provider API

AI / embeddings

EXTERNAL / LOCAL INFRASTRUCTURE INTEGRATIONS

Google OAuth

Identity validation
Auth-service integration

RabbitMQ

Reservation outbox -> notification
Authenticated HTTP fallback

Razorpay TEST

Orders, captures, refunds
Webhook + reconciliation

SMTP provider

Auth account emails
Notification lifecycle / support mail

Gemini Developer API

Chat generation + embeddings
Free-tier provider availability varies

DB ownership is shown inside each service. All eight schemas can share a single local MySQL instance; there are no normal cross-service SQL joins.
Key internal paths: Reservation -> Train/Schedule/PNR; Chatbot -> authorized live APIs; protected services -> Auth session/account validation.
Booking events: Reservation outbox -> RabbitMQ (or internal HTTP) -> Notification inbox -> SMTP. Account mail: Auth -> SMTP.
External integration names identify their service relationships. No Redis, Docker or WSL is required; secrets remain in private environment configuration.

Source: backend/README.md, service configuration and current API clients. Arrows show selected request/event directions; return traffic and every internal endpoint are omitted.